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AI for Bharat Hackathon

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Team Name : Team Meridian

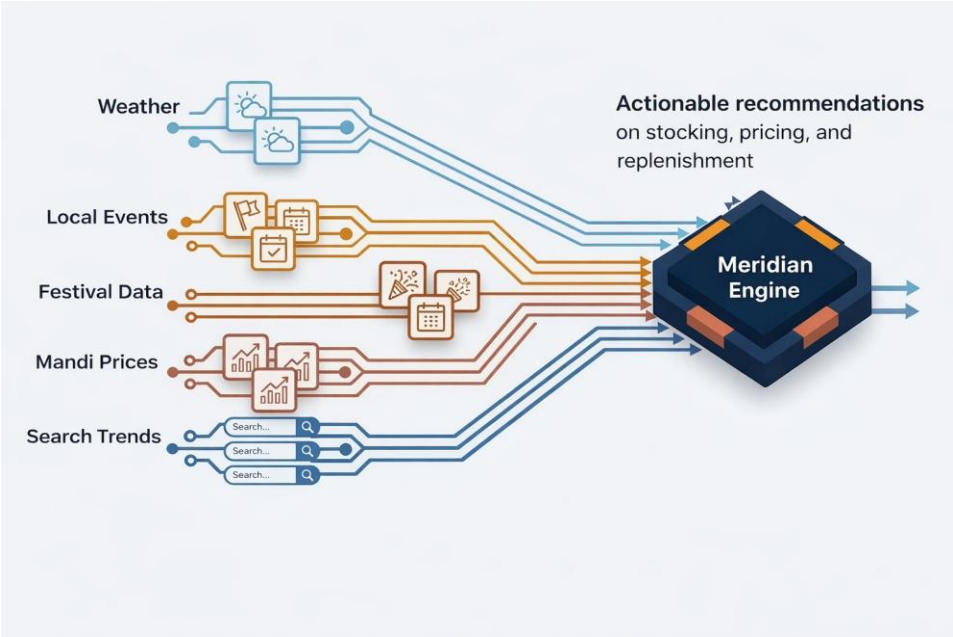
Team Leader Name : Gayatri Srinivasan

Problem Statement : Disconnected supply chains in Indian retail markets lead to significant inventory waste and missed revenue opportunities due to a lack of synchronization between local inventory and global/hyper-local market signals.

Meridian Commerce Engine - The AI strategy copilot for Indian Kirana retail

Disconnected Data to Intelligent Hyperlocal Retail

- ✓ An AI-powered anticipatory commerce system designed specifically for "Bharat.“
- ✓ Transforms traditional inventory management using Virtual Category Manager Copilot
- ✓ Synthesizes real-time hyperlocal signals along with global trends with historical FCMG Data
- ✓ Generates actionable recommendations on stocking, pricing and replenishment



The AI & AWS Value Proposition

Why AI?

Enables dynamic reasoning over unstructured external data (weather impacts, festival spikes), moving far beyond static, hard-coded rules.

How AWS is Used?

A fully serverless orchestration backbone (Lambda, API Gateway, DynamoDB) powering advanced LLM interactions via Bedrock.

What Value does AI Add?

Provides explainable recommendations (Live Reasoning Feed) and conversational access to deep FMCG domain knowledge (RAG).

List of Features

1. Hyperlocal Strategy Generation

Live reasoning feed utilizing Bedrock's tool-use loop to synthesize 4 external APIs into actionable insight cards.

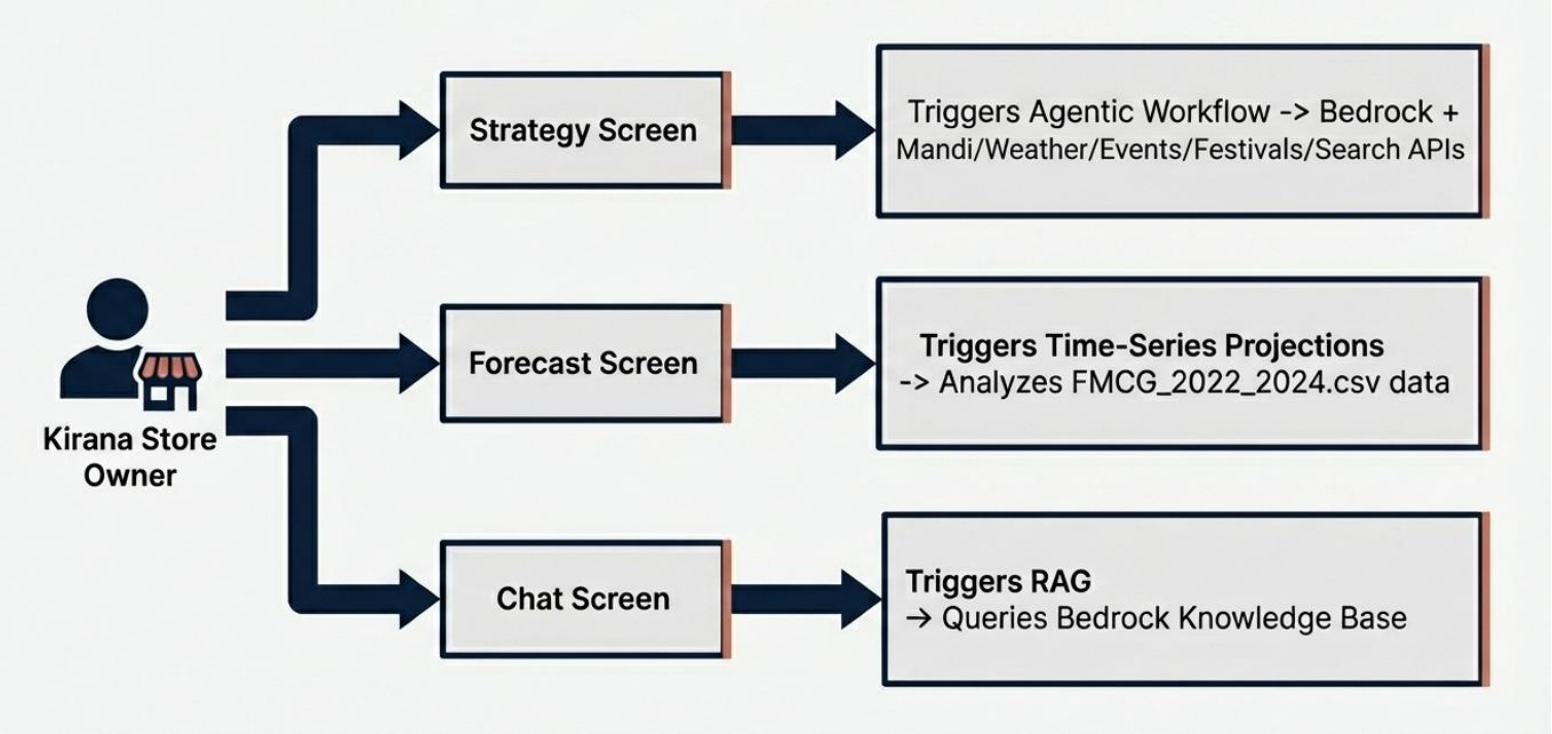
2. Predictive 15-Day Forecasting

Time-series demand projections comparing current data against a 2023 historical baseline to flag stock-to-sales risks.

3. Virtual Category Manager

Conversational RAG interface allowing operators to query FMCG domain knowledge with session continuity.

Use Case/ Process Flow



Mockup of the UI Screens

KIRANA STORE Shree Ganesh Stores | Dombivli East, Maharashtra

Hyperlocal Strategy

Anchor date: 2026-03-08

Live Reasoning Feed

- Executing: get_local_events
- Executing: get_upcoming_festivals
- Executing: get_mandi_prices
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Overview | Hyperlocal Strategy | Predictive Forecasts | Virtual Category Manager

Generate Insights

Insight Cards

FASTING (VRAT) - VARUTHINI EKADASHI
Insight: High demand for Sabudana, Peanuts, Makhana, and Rock Salt.
Action: Stock up on fasting staples and promote them in the store.
Rationale: Varuthini Ekadashi is a fasting day, leading to increased demand for fasting staples.

FESTIVAL - GUDI PADWA (MARATHI NEW YEAR)
Insight: Surge in demand for premium groceries and traditional ingredients.
Action: Increase stock of premium groceries, traditional ingredients, and festive decorations.
Rationale: Gudi Padwa is a major shopping day, with a focus on premium groceries and traditional ingredients.

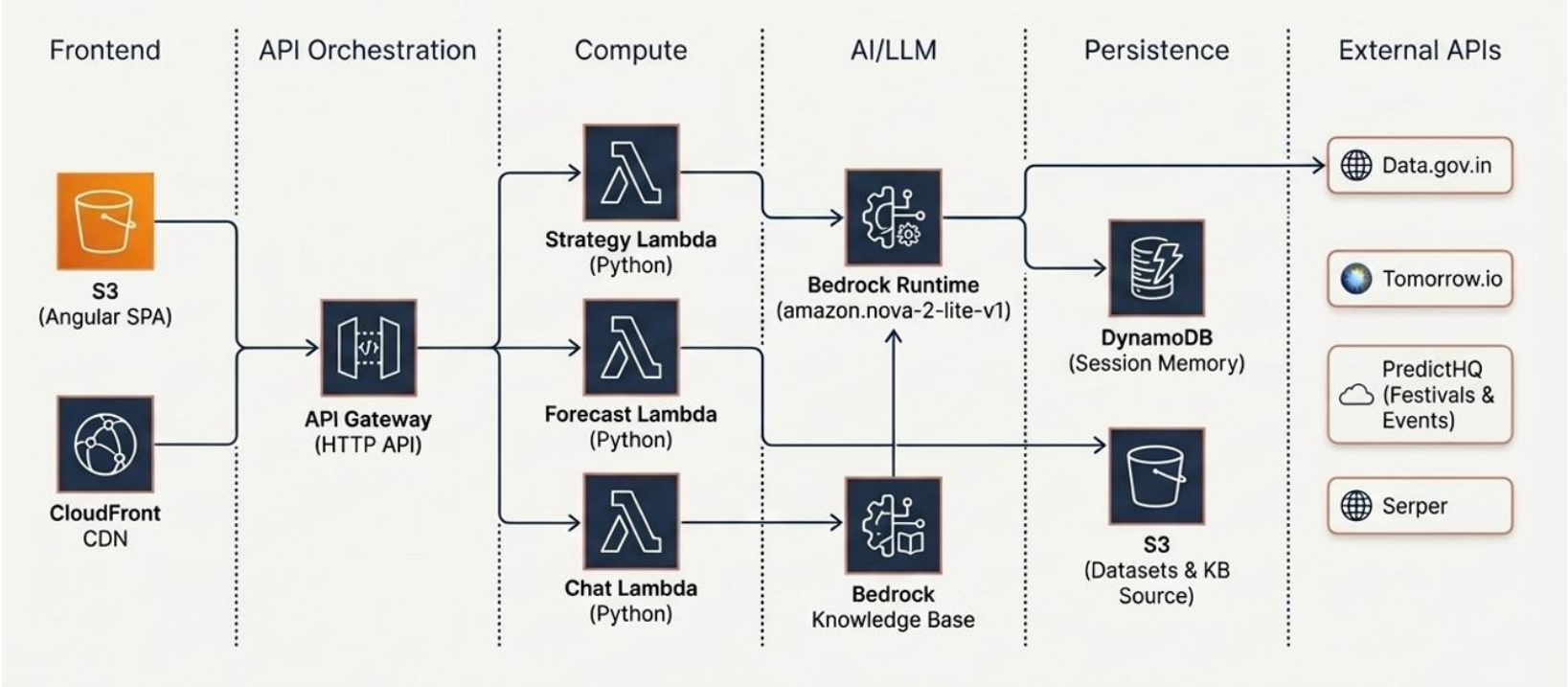
CIVIC - SUNDAY MARKET / WEEKEND
Insight: High footfall in residential clusters like Dombivli.
Action: Promote bulk purchasing and offer discounts on meat and staples.
Rationale: Sunday markets attract high footfall, leading to increased demand for meat and staples.

DAIRY - MILK
Insight: Historically, milk has a high average stock held and total units sold.
Action: Maintain a higher stock level for milk and offer promotions to boost sales.
Rationale: Milk has a consistent demand, and promotions can help increase sales.

UI Focus:
Dark-themed, responsive design built with standalone Angular components.

Key Behavior:
User-triggered insight generation with explicit rationale display for explainability.

Architecture Diagram



Technology Stack

Frontend

- Angular
- TypeScript
- RxJS
- CSS

Backend Compute

- Python
- AWS Lambda
- Amazon API Gateway

AI & Data Persistence

- Amazon Bedrock (Nova-lite)
- Amazon DynamoDB
- Amazon S3

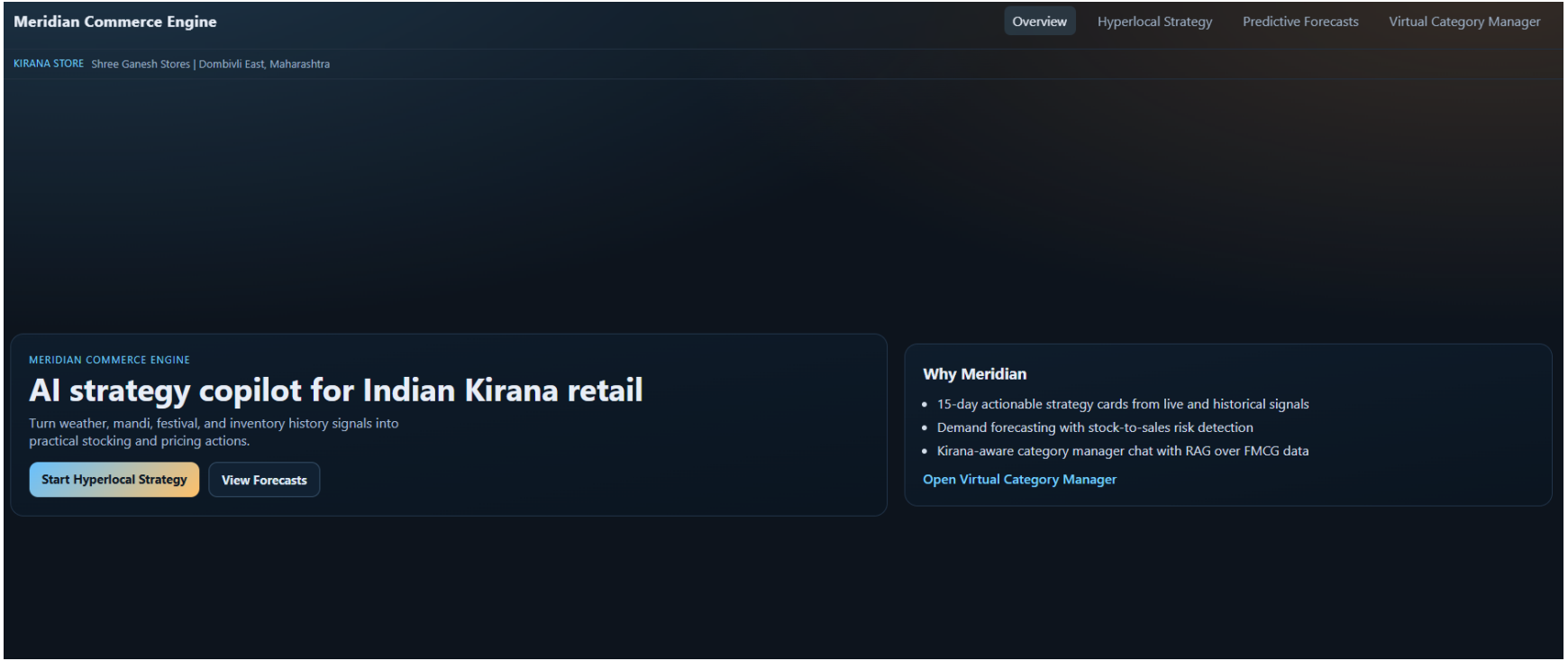
External Signals

- **Tomorrow.io** (Weather)
- **PredictHQ** (Events)
- **Serper** (Search)
- **Data.gov.in** (Mandi)
- **Festivals** (Indian Calendar)

Implementation Costs Estimated

Component Class	Technology	Billing Model
Compute & API	Lambda & API Gateway	Pay-per-use scaling
AI Intelligence	Bedrock Converse & RAG	On-demand token pricing
Hosting & Storage	S3 & DynamoDB	Low-cost static hosting & On-demand tables

Prototype Snapshots – Home Screen



Prototype Snapshots – Screen 1

Meridian Commerce Engine

Overview | Hyperlocal Strategy | Predictive Forecasts | Virtual Category Manager

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Hyperlocal Strategy

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Insight Cards

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Prototype Snapshots – Screen 2

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Predictive Forecasts

Run 15-Day Forecast

Model: Weekday-seasonality + short-term trend (TFT-like lightweight baseline)

Demand Projection (2026) vs 2023 Baseline Window

Date	Category	Predicted Units
2026-03-09	Juice	83
2026-03-09	Milk	756
2026-03-09	ReadyMeal	810
2026-03-09	SnackBar	723
2026-03-09	Yogurt	1,327
2026-03-10	Juice	82
2026-03-10	Milk	740
2026-03-10	ReadyMeal	810
2026-03-10	SnackBar	710
2026-03-10	Yogurt	1,318
2026-03-11	Juice	85
2026-03-11	Milk	738
2026-03-11	ReadyMeal	811

Prototype Snapshots – Screen 3

Meridian Commerce Engine

Overview | Hyperlocal Strategy | Predictive Forecasts | Virtual Category Manager

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Virtual Category Manager

Conversational RAG over FMCG history using Bedrock Knowledge Bases.

You

Which categories need price increase due to mandi/weather signals?

Manager AI

There is no direct information in the provided search results about mandi/weather signals affecting prices.

- The data only includes sales records with prices, regions, and pack types.
- No external factors like mandi rates or weather impacts are mentioned.
- Focus on categories with low stock_available or high units_sold to decide price adjustments.
- Consider pack_type differences (Single vs Carton) for balanced pricing.
- Check promotion_flag to avoid conflicting discounts during price increases.

No specific categories are indicated for price increases based on the given data.

Ask an inventory question

Send

Quick prompts

How much milk should I stock for next 10 days? | Which SKUs have highest stockout risk this week? | What should I reorder first for promoted carton packs? | Show quick reorder plan by region for beverages. | Which categories need price increase due to mandi/weather signals?

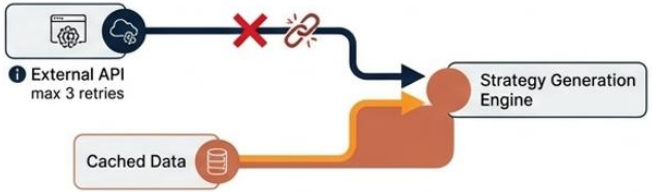
Prototype Performance and Benchmarking

Dual Testing Approach: Unit testing paired with Property-based testing (hypothesis library) for 12 strict correctness properties.

Performance Limits:

Strict Lambda timeout management enforced.

Graceful Degradation: Circuit breakers implemented for external APIs (max 3 retries); system continues strategy generation with available cached data if an API fails.



Future Plans

Current Implementation – Hyperlocal Intelligence, 15-Day Forecasting, Conversational RAG

Future Plans:

Supplier Order Automation: The system could be expanded to automatically trigger replenishment orders with suppliers based on the AI's predictive forecasts and inventory recommendations, streamlining the supply chain

Real-Time Inventory Tracking & System Integration: While the current engine relies on historical FMCG data, a future plan could involve integrating with software-based inventory management systems to track and update stock levels in real time

Prototype Assets

Github Link: https://github.com/gayatri-01/meridian-commerce-engine-copilot

Demo Video Link: https://drive.google.com/file/d/1-v_7n4NEB3352hgK2Y1aNyj7ALewlmif/view?usp=sharing

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Thank You